

Modelling cycling route choice using a map-based public participatory survey in a regional city

Sapan Tiwari^{1,*}, Afshin Jafari¹, Steve Pemberton¹, Alan Both¹, Dhirendra Singh²,
Ryan Turner¹ and Chris De Gruyter¹

¹*Centre for Urban Research, RMIT University, Melbourne, Australia*

²*CSIRO's Data61, Private Bag 10, Clayton South VIC 3169, Australia*

**Corresponding author, Email address: sapan.tiwari@rmit.edu.au, Postal address: RMIT University, GPO Box 2476, Melbourne VIC 3001*

Abstract

Riding a bicycle, particularly for transport purposes, offers substantial environmental and health benefits. Suitable bicycle infrastructure is crucial for promoting bicycle use, but remains scarce and fragmented in Australia, particularly in regional cities. In this context, understanding cyclists' route choices is a key input in informed infrastructure planning to increase cycling mode share. This study uses data from a map-based public participatory route survey to spatially capture bicycle route choice in the regional city of Greater Bendigo, Australia. The model incorporates attributes such as level of traffic stress (LTS), slope, tree canopy cover, intersection density, and route directness, along with an adjustment to account for overlap among alternative routes. The results show that cyclists strongly prefer routes with the lowest traffic stress, lower gradients, and greater network connectivity, while avoiding circuitous paths. Cyclists are also more likely to choose routes that are distinct and share fewer common segments with alternative routes. Segmentation by gender and age reveals notable behavioural differences: female cyclists are considerably more sensitive to traffic stress and slope, prioritising safety and comfort, whereas male cyclists exhibit greater tolerance to stress and prioritise route efficiency. Middle-aged cyclists (40–60 years) exhibit the strongest aversion to stressful routes, while younger cyclists (18–40 years) demonstrate greater flexibility in route choice. These findings highlight the importance of age and gender considerations in bicycle infrastructure planning as well as prioritising low-stress, well-connected, and direct cycling corridors. The findings also highlight the value of map-based public participatory surveys as a cost-effective means of collecting route data in regional settings with small populations.

Keywords— Bicycle, Level of traffic stress, Path Size Logit, Route choice model, PPGIS

1 Introduction

Bicycle riding offers significant environmental and health benefits. When used as an alternative to motorised transport, it helps reduce road congestion, air pollution and carbon emissions while improving physical activity (Brand et al., 2021). As a result, many cities are investing in efforts to increase bicycle use as part of broader sustainability and climate adaptation strategies. Delivering safe, equitable, and efficient cycling networks requires an understanding of how cyclists choose their routes (Łukawska, 2024). Robust route choice models are central to this task because they help quantify how riders prioritise distance, safety, infrastructure quality, and other attributes in their routing decisions, while also providing an evidence base for network design, the appraisal of cycling investments, and the evaluation of policy scenarios.

Recent studies commonly use GPS trajectory data to estimate bicycle route choice models and to examine how riders select their routes in practice (Meister et al., 2024; Lilasathapornkit et al., 2025). GPS traces provide detailed information on where, when, how long, and how far people ride, but are costly to collect, often have low participation rates, and capture behaviour only during the monitoring period. These challenges are more pronounced in areas with low cycling mode shares, such as regional cities with smaller populations.

Public Participatory Geographic Information Systems (PPGIS), as a digital participatory mapping method, enables public involvement in the collection of non-expert spatial data (Brown and Kyttä, 2014). Using web-based tools, such as Mapptionaire, respondents can map their habitual routes and report perceptions of route quality at relatively low cost, while also providing flexibility for use in regional settings. Unlike GPS data, however, PPGIS data does not include temporal information such as travel times. Despite this limitation, PPGIS provides a rich and promising data source for route choice modelling (Alattar et al., 2021) because it represents a self-reported, distilled summary of a rider’s typical route. Additionally, it provides information on *perceived* route qualities that cannot be derived from GPS trajectory data alone.

A key challenge is that the performance of route choice models depends critically on the quality of the constructed choice sets (Rieser-Schüssler et al., 2013; Tahlyan and Pinjari, 2020). Traditional approaches such as the Multinomial Logit (MNL) assume independence among alternatives, which is problematic when routes share substantial segments and therefore have correlated utilities. Advanced formulations such as the Path Size Logit Path Size Logit (PSL) (Ben-Akiva et al., 2004) and its mixed extension Mixed Path Size Logit (MPSL) account for route overlap and unobserved heterogeneity while retaining a tractable closed form. Parallel advances have been made in choice set generation, including attribute-based labelling of routes by safety or gradient (Ben-Akiva et al., 1984; Broach et al., 2010), network-based methods such as Breadth First Search with link elimination (Rieser-Schüssler et al., 2013; Tahlyan and Pinjari, 2020) and data-driven techniques that construct choice sets directly from observed paths (Ton et al., 2018).

While most prior research on modelling cycling route choice has focused on large metropolitan areas with dense, well-connected cycling networks (Broach et al., 2012; Prato et al., 2018; Meister et al., 2023; Lilasathapornkit et al., 2025; Kavta et al., 2025), research in regional cities remains very limited, despite their distinctive travel environments. In addition to differences in network density, infrastructure quality, and traffic composition, regional cities also have smaller and more dispersed populations with lower cycling mode shares. These demographic and behavioural differences further limit the direct transferability of modelling frameworks developed for metropolitan contexts. Furthermore, the integration of PPGIS based route mapping with advanced route choice models has seen limited application to date.

This study addresses these gaps through a case study of the City of Greater Bendigo (CoGB), a regional city in Victoria, Australia. It develops and estimates bicycle route-choice models using PSL and MPSL formulations,

based on a public participatory route-mapping survey in which 278 cyclists indicated their usual riding routes. The stated routes are matched to a detailed bicycle network derived from OpenStreetMap, and a hybrid choice-set generation framework is implemented that combines attribute-based, network-based, and stochastic techniques to construct comprehensive and sensible alternative routes anchored to the surveyed paths. Using these data and models, the paper quantifies cyclists' sensitivities to infrastructure, slope, detour, and traffic stress, and explores systematic differences in preferences across gender and age groups. In doing so, we showcase the utility of PPGIS as a low-cost survey-based approach to support route choice modelling, provide a route choice model for regional cities, and also advance a practical hybrid framework for choice set generation in data-scarce contexts. In turn, the findings provide evidence of the relative importance of various attributes influencing cyclists' route choice – including differences by gender, age, and trip purpose – which can inform future planning of bicycle infrastructure in regional cities to accommodate a range of needs and abilities.

2 Literature review

2.1 Modelling approaches for cycling route choice

Research on cycling route choice behaviour has expanded rapidly in recent years, supported by advances in behavioural modelling and richer representations of cycling networks. Early studies commonly adopted the MNL framework to model cyclists' decisions among discrete route alternatives (Broach et al., 2012; Menghini et al., 2010). However, when route alternatives share substantial portions of their paths, the model's assumption of independence among alternatives is violated. Such overlap generates correlation in unobserved utilities, which can bias parameter estimates and reduce behavioural realism.

To address these limitations, a range of correction-term logit models that add a similarity measure to the deterministic utility were developed. The most prominent examples are the PSL (Ben-Akiva et al., 2004) and the C-Logit (Cascetta et al., 1996), both of which incorporate similarity terms into the deterministic utility to adjust for infrastructure links shared with other travel modes. These models retain the closed-form structure of the logit models, making them computationally powerful while providing more realistic behavioural responses. PSL has been widely adopted for cycling, with empirical studies conducted in Portland, Zurich, San Francisco, Copenhagen, Melbourne, and other cities (Broach et al., 2012; Menghini et al., 2010; Casello and Usyukov, 2014; Ton et al., 2017; Lilasathapornkit et al., 2025; Prato et al., 2018; Meister et al., 2024, 2023).

Beyond correcting for overlap, a second line of work extends route-choice models to incorporate preference heterogeneity, recognising that cyclists vary substantially in their sensitivity to attributes such as distance, infrastructure, gradient, and traffic conditions (Łukawska, 2024). Mixed-coefficient (random-parameter) logit formulations, including the MPSL, allow coefficients to vary across individuals while retaining a Path Size (PS) correction term, which adjusts route utilities according to the degree of overlap with other alternatives, ensuring that highly similar routes are not treated as independent choices (Hensher and Greene, 2003; Sener et al., 2009; Lilasathapornkit et al., 2025). Studies in Portland, Zurich and Melbourne have shown that mixed formulations provide richer behavioural insights and help explain systematic heterogeneity related to gender, age and e-bike usage (Broach et al., 2012; Meister et al., 2023; Lilasathapornkit et al., 2025).

Beyond path-based approaches, link-based dynamic formulations such as the Recursive Logit (RL) model have been proposed (Fosgerau et al., 2013). RL models represent route choice as a sequence of link-level decisions and avoid the need for explicit choice-set generation. Extensions incorporating spatial lag terms and network-dependent utilities have further enhanced their behavioural realism (van Nijen et al., 2024). Despite these advances, path-

based models such as PSL and MPSL remain attractive for planning applications because they offer a transparent interpretation of how cyclists value specific route attributes.

Alongside these modelling advances, the type of data used to estimate route-choice models has also evolved and plays an essential role in the modelling. Most empirical studies rely on GPS-based revealed-preference data, trajectories collected via loggers, smartphones, or crowdsourced platforms (Broach et al., 2012; Menghini et al., 2010; Prato et al., 2018; Meister et al., 2023). These datasets provide detailed observations of realised routes, but are costly to collect and often remain limited to large metropolitan areas with high cycling mode shares. Conversely, crowdsourced datasets such as Strava can provide larger sample sizes but tend to overrepresent confident or recreational riders (van Nijen et al., 2024). Recent work by Lilasathapornkit et al. (2025) demonstrates the use of smartphone-based GPS tracking in Greater Melbourne, enabling participants to record their actual routes in real time. However, in regional or lower-density settings, such GPS-based data are often sparse, limiting the robustness of model estimation.

Given these limitations, particularly in areas with low cycling volumes, map-based revealed-route surveys have emerged as a practical alternative (Brown and Kytä, 2014; Alattar et al., 2021). These surveys allow respondents to digitally trace their habitual routes and provide contextual information on perceived route quality. While lacking temporal detail compared with GPS traces, they offer a feasible and cost-effective means of collecting spatially explicit route data in settings where GPS samples may be sparse.

Taken together, the literature highlights the suitability of PSL and MPSL models; however, their application is often limited by the scarcity of GPS data outside major cities. This study addresses these limitations by employing a map-based revealed-route survey, offering a practical and innovative data source for regional contexts. Integrated with a hybrid choice-set generation approach, it enables the estimation of cyclists' route preferences in a regional setting.

2.2 Factors associated with cyclists' route choice behaviour

Studies of cycling route choice use diverse sets of attributes to explain how riders select their routes (Łukawska, 2024). Across different study areas, research consistently shows that cyclists do not simply minimise distance but instead prioritise routes that reduce exposure to traffic, enhance safety, and improve overall comfort (Broach et al., 2012; Menghini et al., 2010; Casello and Usyukov, 2014; Bernardi et al., 2018; Ghanayim and Bekhor, 2018; de Jong et al., 2023). These preferences are influenced by physical infrastructure, road environment, topography and surface quality, as well as broader environmental and land-use characteristics (Łukawska, 2024).

Literature suggests that infrastructure quality consistently emerges as the strongest predictor of route choice. Empirical studies in cities such as Portland, Zurich, and Copenhagen found that cyclists are willing to accept longer travel distances to use protected lanes, off-street paths, and other physically separated facilities (Broach et al., 2012; Menghini et al., 2010; Prato et al., 2018). Well-maintained and continuous facilities substantially increase route attractiveness (Bernardi et al., 2018; Casello and Usyukov, 2014), while painted lanes or discontinuous infrastructure segments can deter use in favour of quieter residential streets with lower traffic volumes (Fitch and Handy, 2020; de Jong et al., 2023; Łukawska, 2024). The continuity of facilities across bridges and intersections is also important for maintaining perceived safety and network coherence (Prato et al., 2018). These findings align closely with the Level of Traffic Stress (LTS) framework, which classifies cycling environments based on infrastructure and traffic characteristics. Attributes such as the proportion of road segments with different levels of traffic stress, central to the present study, capture key variations in cycling conditions identified in past literature.

Traffic exposure and intersection design are also influential. Multiple studies have identified that higher traffic volumes, wider roads, and greater motor-vehicle speeds consistently reduced route attractiveness (Menghini et al.,

2010; Broach et al., 2012; Fitch and Handy, 2020; de Jong et al., 2023). Cyclists tend to prefer local streets with lower speed limits and fewer lanes, whereas complex junctions, unsignalised crossings, and frequent turns tend to be perceived as less safe and more stressful (van Nijen et al., 2024). Some studies have observed that intersections controlled by traffic signals increase perceived safety but introduce delays, illustrating the behavioural trade-off between safety and efficiency (Broach et al., 2012; de Jong et al., 2023).

Topography and surface quality also strongly influence cyclists’ route choice. Avoidance of steep gradients is among the most consistent findings in the literature, with cyclists choosing flatter routes even when this introduces additional distance (Menghini et al., 2010; Łukawska, 2024; van Nijen et al., 2024). The introduction of e-bikes has partly mitigated this constraint, as e-bike users display reduced slope sensitivity (Dane et al., 2019; Lilasathapornkit et al., 2025). Road surface quality is another critical factor, with cyclists favouring smooth, paved routes and dedicated paths, and avoiding rough or unsealed surfaces (Bernardi et al., 2018; Fitch and Handy, 2020; de Jong et al., 2023).

Beyond physical design, environmental and land-use attributes also influence cyclists’ route choice. Routes featuring tree canopy cover, greenery or park interfaces can provide shade, reduce heat exposure and enhance perceived attractiveness (de Jong et al., 2023; Łukawska, 2024). However, heavy commercial corridors may be avoided due to pedestrian congestion and delivery vehicle activity, despite offering direct connections (Bernardi et al., 2018; van Nijen et al., 2024). These findings underscore the importance of including tree canopy cover in the present study to capture aesthetics that are often overlooked in traditional network analyses.

Cyclists also consider route coherence and complexity. Measures such as the number of turns, intersections, or deviations from the shortest path are widely used to characterise network legibility and navigational burden. Routes with fewer interruptions, clearer line-of-sight and greater continuity tend to be preferred (Broach et al., 2012; Menghini et al., 2010; Prato et al., 2018; de Jong et al., 2023). The detour ratio included in this study, therefore, captures the behavioural trade-offs between efficiency and safety commonly observed in previous work.

Taken together, these studies highlight that cyclists balance safety, comfort, environment and efficiency within the constraints of the built environment and their own capabilities. However, as noted by Łukawska (2024) and de Jong et al. (2023), the majority of this empirical evidence comes from large metropolitan areas with dense GPS datasets. In such contexts, cyclists can often access low-stress or high-quality facilities without incurring substantial detours. Regional and medium-sized cities remain understudied despite their distinctive network structures, topographic profiles and traffic compositions. The present study addresses this gap by using a map-based revealed-route survey to quantify how infrastructure quality (via LTS measures), slope, canopy cover, route complexity (intersection density) and indirectness (detour ratio) influence cyclists’ commuting route choices in a regional context.

2.3 Choice set generation

The generation of choice sets is a fundamental step in route choice modelling, as the credibility and behavioural realism of estimated models depend heavily on how alternative routes are constructed (Halldórsdóttir et al., 2014; Flötteröd and Bierlaire, 2013; Meister et al., 2024). The literature offers a wide range of techniques that differ in their underlying assumptions, computational demands and ability to generate diverse, plausible alternatives (Ton et al., 2018).

Deterministic methods are among the most common and widely used categories. These include techniques such as k -shortest paths, link elimination, labelling, and link penalty techniques, all of which modify the shortest path under different cost structures (Łukawska, 2024). The k -path algorithms are straightforward to implement but often produce alternatives with high spatial overlap (Ton et al., 2018). Link elimination approaches, such as Breadth-

First Search on Link Elimination (BFSLE), iteratively remove links from previously generated paths to uncover new feasible alternatives and have been applied in both cycling and freight contexts (Rieser-Schüssler et al., 2013; Tahlyan and Pinjari, 2020; Meister et al., 2024). Labelling methods take a different approach and reflect the heterogeneity of cyclists' preferences by optimising routes with respect to specific attributes, such as infrastructure type, slope, or number of intersections (Ben-Akiva et al., 1984; Ton et al., 2018). Labelling methods can capture the role of key route features but may overlook alternatives that are not aligned with predefined labels.

Stochastic methods aim to introduce behavioural variability by randomising the impedances used in shortest path calculations. Methods such as the doubly stochastic generation function (DSGF) (Halldórsdóttir et al., 2014) and Metropolis–Hastings (MH) sampling (Flötteröd and Bierlaire, 2013; Sobhani et al., 2019) use randomised impedance functions or probabilistic acceptance rules to generate diverse alternatives. These approaches can approximate the variability observed in real-world cycling but are computationally intensive and require well-specified distributions for attribute sensitivities.

A common concern across all methods above is the high degree of overlap among generated alternatives. To address this, several studies impose explicit overlap thresholds or use similarity indices to filter alternatives (Tahlyan and Pinjari, 2020; Łukawska et al., 2023). Despite such refinements, detailed information about parameter choices or penalty schedules is often lacking, making replication difficult (Ton et al., 2018). Notable exceptions include Ton et al. (2018) and Meister et al. (2023), who provide transparent descriptions of penalty functions and algorithmic settings.

Recent work, therefore, increasingly combines multiple techniques into hybrid frameworks, recognising that no single method performs best across all contexts. Deterministic algorithms provide structure and reproducibility, stochastic methods enhance behavioural diversity, and data-driven approaches help anchor alternatives in observed behaviour. This insight is particularly relevant in regional settings, where data scarcity limits the construction of purely GPS-based or data-driven choice sets.

In line with these developments, the present study employs a hybrid choice-set generation framework grounded in survey-reported routes. Attribute-based labelling, link elimination, and stochastic variation are combined to produce diverse yet plausible alternatives that reflect cycling environments in regional settings. This approach ensures that alternatives differ meaningfully from the chosen route while capturing the range of infrastructure, slope, stress and complexity conditions present in the network. By integrating deterministic, behavioural and survey-anchored elements, the resulting choice sets support robust estimation of the PSL and MPSL models in a regional context.

3 Methods

In line with recent developments in the literature, this study employs a hybrid choice-set generation framework grounded in survey-reported routes. Attribute-based labelling, link elimination and stochastic variation are combined to produce diverse yet plausible alternatives that reflect the feasible cycling environment of the chosen area. This approach ensures that alternatives differ meaningfully from the chosen route while capturing the range of infrastructure, slope, stress, and complexity conditions present in the network. By integrating deterministic, behavioural and survey-anchored elements, the resulting choice sets support robust estimation of the PSL and MPSL models in a regional context.

3.1 Case study

The case study for this research is the City of Greater Bendigo (CoGB), a major regional municipality with a population of approximately 125,000, situated approximately 150 km northwest of Melbourne, Victoria. CoGB is a medium-sized regional centre that combines an urban core with surrounding peri-urban and rural areas (Chen, 2016) and has been identified as one of Victoria's key regional growth centres (Victorian Government, Department of Transport and Planning, 2014). The transport network in CoGB combines a grid layout in the historic core with radial and curvilinear patterns in the suburbs, linked by major arterials. The spatial distribution of existing cycling infrastructure is shown in Figure 1, which includes both off- and on-road bicycle infrastructure connecting outer suburbs to the central city area. Numerous off-road paths exist inside the nature reserves in peri-urban areas. The CoGB provides an interesting setting for examining cyclist route choice, with lower traffic volumes and shorter travel distances leading to lower traffic stress than in metropolitan areas, yet facing elevated deterrents such as discontinuous infrastructure and variable topography.

3.2 Survey design and data collection

Data were collected using an online map-based survey administered through the *Maptionnaire* platform. The survey targeted people who ride a bicycle in CoGB and was open from 24 March to 25 April 2024. The survey link was shared through local bicycle user groups and CoGB social media channels. A small monetary incentive was included, as the chance of winning one of three gift cards valued at \$200 (AUD). Participation was voluntary, but eligibility was restricted to those who were at least 18 years old and had cycled in the CoGB at least once in the previous year. In total, 284 people responded to the survey. Of these, 6 did not meet the eligibility criteria, and another 3 did not satisfactorily complete the survey and were excluded, leaving 275 eligible respondents.

The survey was structured into three parts: sociodemographic characteristics, cycling experiences, and cycling preferences. The sociodemographic section asked about age, gender, household structure, number of cars, and income. In the cycling experience section, respondents used an interactive map of CoGB to draw their preferred cycling routes, for example, to work, education, shops, family and friends, public transport, or other destinations, and were asked to map at least one route. The survey interface and route-drawing task are illustrated in Appendix Appendix A:. The final section asked about cycling preferences and priorities, including what changes or improvements they would like to see to make cycling easier and more attractive in CoGB.

In total, 454 route responses were recorded, as some respondents added more than one route. Some of these routes were later rejected by the research team as not useful for mode choice analysis. Of the 454 survey responses, six had not drawn a route, seven had drawn one so short that its start and end points mapped to the same network node, and 13 had drawn a circular route that returned to its starting point (which resulted in the route not being valid as any attempt to find alternative routes between the identical start and end point would fail). Another 169 routes with the tagged purpose "to other destinations" were also removed, as these primarily represent recreational travel, which is more discretionary and involves different decision-making processes and route choice behaviour (Piras et al., 2021). This resulted in a final sample of 259 routes used for modelling route choice behaviour.

Table 1 provides a summary of the socio-demographic characteristics of the survey participants. As shown, there were slightly more male participants (54%) than female (45%). Approximately 72% of participants were aged over 40 years, with a notable proportion aged over 60 (27%). Most participants (77%) were living with a partner as a couple, and almost half (46%) were living with children. Most participants (88%) reported a tertiary qualification of some kind. The most common income category was \$45,001 to \$135,000 (36% of participants). Approximately 50% of households had two cars, whereas only approximately 18% had more than two cars. Households with multiple

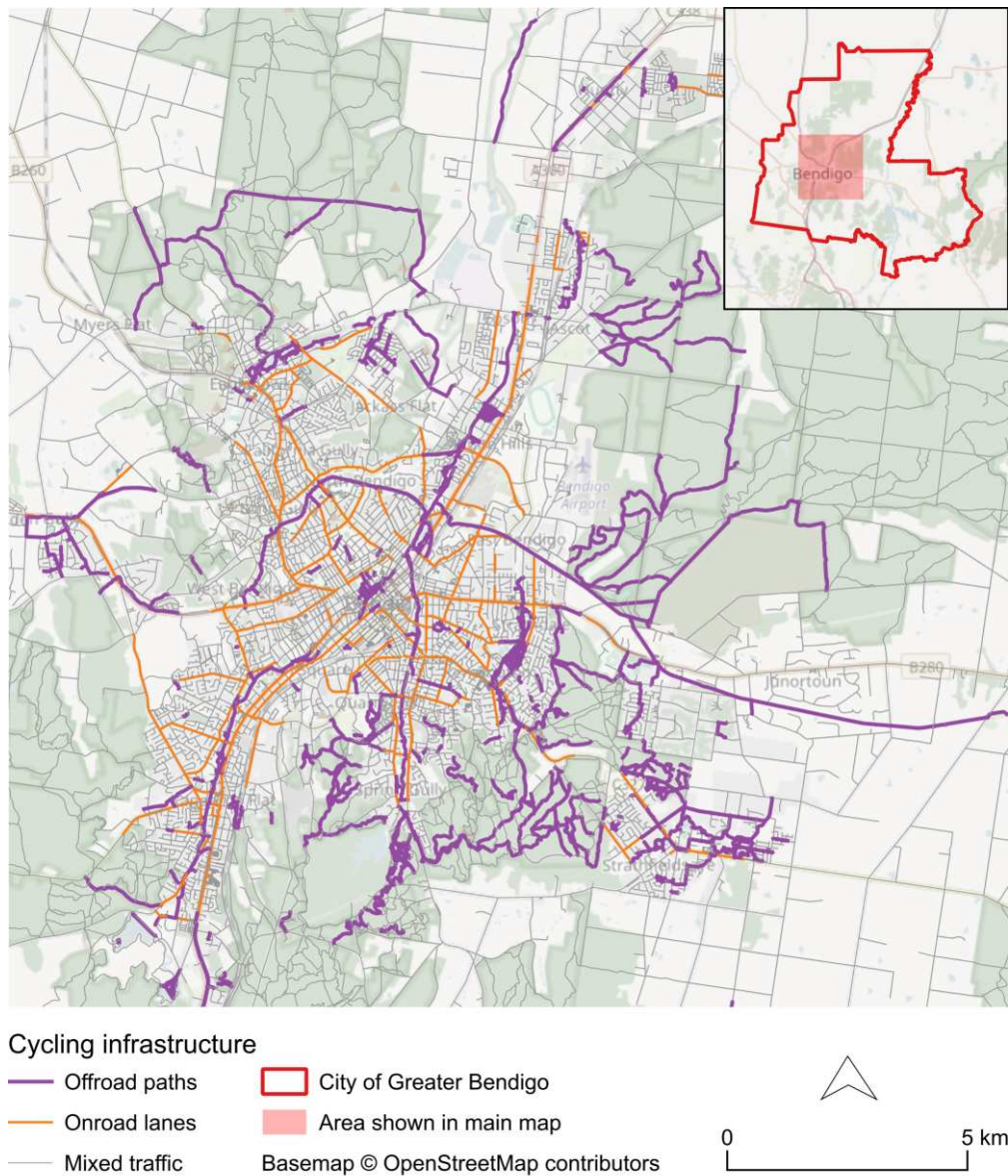


Figure 1: Bicycle infrastructure network in Greater Bendigo region

bicycles were common: 21% owned two bicycles, and 70% owned more than two bicycles.

Each participant reported one or more bicycle trips in the survey. Table 2 provides an overview of bicycle trip purposes reported by the survey participants. The most common trip purpose was 'other' (38%), followed by 'work/commute' (27%). Participants who answered 'other' were asked to provide a description, and, although a variety of purposes were stated, most related to recreation.

Table 1: Summary of respondents (N=275) characteristics

Characteristic	n	%
Gender		
Woman/Female	123	44.7
Man/Male	149	54.2
Other / prefer not to say	3	1.1
Age group		
18–40 years	76	28.1
40–60 years	121	44.8
60 years or more	73	27.0
Living arrangements		
Couple with children	115	42.6
Couple, no children	94	34.8
Single person, no children	39	14.4
Lone parent with child/children	8	3.0
Other / prefer not to say	14	5.2
Education		
High school or less	28	10.5
Diploma/certificate	67	25.2
Undergraduate	86	32.3
Postgraduate	82	30.8
Other	3	1.1
Annual household income (AUD)		
\$45,000 or less	28	10.8
\$45,001–\$135,000	94	36.4
\$135,001–\$190,000	47	18.2
\$190,000 or more	42	16.3
Prefer not to say	47	18.2
Number of cars in household		
0	6	2.2
1	80	29.9
2	133	49.6
3 or more	49	18.3
Number of bicycles in household		
0	2	0.7
1	20	7.4
2	58	21.5
3 or more	190	70.4

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Note: The sum of *n* may not equal 275 within each section, as some participants did not respond to all questions.

Table 2: Trip purpose reported by survey participants

Trip purpose	n	%
To work/commute	124	27.3
To college/university/classes	13	2.9
To shops/tasks/errands	95	20.9
To visit family/friends	37	8.1
To public transport stop/station	13	2.9
To other destinations	172	37.9
Total	454	100

3.3 Data processing

A network of the existing road and bicycle paths in and around CoGB was derived from OpenStreetMap, using methods based on those described in [Jafari et al. \(2022\)](#). Survey routes, i.e., routes drawn by survey participants, were then mapped to the network. To achieve this, both vertices of every route line drawn by the survey participant were matched to the nearest corresponding nodes in the network model, and then Dijkstra’s algorithm was used to find the least-cost path between those nodes. This allowed every line segment drawn by the participant on the map to be transferred to a corresponding sequence of line segments in the model representation of the underlying road/cycling infrastructure.

It is worth noting that the process of mapping a drawn line segment to modelled network nodes can introduce errors. One reason is that a drawn line segment will not necessarily line up with road segments in the underlying model of the physical infrastructure; for instance, if the line is drawn where no infrastructure exists, or the start/end locations of the drawn line do not overlap with nodes/intersections in the model. An additional nuance is that nodes in the model can be artificial, for instance, where a long road segment is modelled as a sequence of segments of maximum length $\mathcal{L}$ (set to 500m). Here $\mathcal{L}$ influences the available “snap points” that the start/end of a drawn line segment could be mapped to, and dictates the maximum snap distance, being $\mathcal{L}/2$, whereby if either vertex of a drawn line segment is more than this distance away from a modelled node, it may indicate a poorly drawn line or represent erroneous input. For these reasons, additional sanity checks were conducted to visually verify if a mapped route matched the intended drawn route. This check did not result in any routes being discarded.

The least-cost path was calculated using the length of each link as its cost. However, as the network sometimes contains parallel links joining the same pair of nodes, for example, a separated cycleway or footpath beside a road, it was assumed in these circumstances that participants had used cycleways in preference to roads, and roads in preference to footpaths. This was achieved by increasing the cost of links unavailable to cyclists (e.g., footpaths) by 15% and decreasing the cost of links representing cycling infrastructure (e.g., bicycle lanes or separated cycling paths) by 10%.

3.4 Choice set generation

The alternative routes were generated to ensure diversity while remaining behaviourally relevant. Key route attributes known to influence cyclists’ route choices were first identified from the literature, and choice-set generation methods were designed to introduce variation along these dimensions. Accordingly, plausible alternative routes for each

survey route were constructed using a combination of three methods (each described below):

- i) BFSLE (Rieser-Schüssler et al., 2013; Ton et al., 2018): 5 alternative routes. In this method, links making up the original least-cost path are successively eliminated from the network, forcing the computation of new least-cost paths.
- ii) Random Weight Perturbation (RWP): 5 alternative routes. In this method, the weights attributed to network links are randomly changed so that new least-cost paths are generated.
- iii) Preferred attributes or labelling (Ben-Akiva et al., 2004; Ton et al., 2018): 6 alternative routes. This method alters the weights of network links in order to favour the selection of routes with particular preferred attributes.

The methods were implemented in R, using the RPostgres library to query the network links represented in a postgres database with the `pgrouting` extension to find least-cost paths using Dijkstra's algorithm.

Initially, the shortest route was selected as the least-cost path between the start and end points of the survey route, with length as the cost. This was followed by the selection of up to 15 BFSLE routes and up to 15 RWP routes. From each candidate pool, the five most distinct routes satisfying the commonality and distance criteria were selected and added to the final choice set for the survey route.

The BFSLE routes for a survey route were selected by taking the shortest path between its start and end points, then successively eliminating individual links of that shortest path from the network and recomputing the shortest path. If the process of eliminating links one at a time (level 1) was not sufficient to find the target number of 15 routes that passed the commonality and distance tests described below, the search continued by eliminating another link from each of the level 1 shortest paths (level 2), and then continuing to subsequent levels if necessary. Within each level, the order of link elimination was determined at random. The process continued until either 15 routes had been found or 1000 route-finding attempts had been made.

A BFSLE route selected using this process was only added to the choice set for its survey route if it passed a commonality test to ensure that it differed from all previously-selected alternative routes in the choice set by at least 20%, and a distance test to ensure that its length was not more than a maximum of twice the length of the shortest route. Commonality was calculated between any two routes a and b using the equation $C_{ab} = L_{ab} / \sqrt{L_a L_b}$, where C_{ab} is the commonality between routes a and b , L_{ab} is the length of the shared portion between the two routes, and L_a and L_b are the lengths of routes a and b , respectively (Tahlyan and Pinjari, 2020). A route was not added to the choice set unless (1) its commonality was 0.8 or less for all previously-selected routes, thereby achieving at least 20% difference, and (2) its length did not exceed twice the length of the shortest route.

The RWP routes were selected by randomly multiplying the length of each link in the network by 1, 2 or 3, and using those values as the cost to calculate a new least-cost path for each survey route. Each route selected in this way was added to the choice set for its survey route if it passed the commonality test relative to each previously-selected alternative route, and if it passed the distance test. Then, the randomised values were recalculated, and a new set of least-cost paths was computed. This process continued for each survey route until either the target of 15 alternative routes had been added to the choice set or 1000 options had been tried.

Where the targets of 15 BFSLE routes and 15 RWP routes were not met, this was generally because the distance between start and end points of the survey route was too short to generate meaningful alternative paths from nearby links.

Preferred attribute routes were each selected to favour a specific attribute as shown in Table 3. The preferred attribute routes, other than the shortest route, were selected from routes first identified using the BFSLE and RWP methods, as described. Following the selection of the shortest BFSLE and RWP routes for each survey route, scores were calculated for each of the attributes shown in Table 3 other than 'short'. In each case, the score was calculated

as $L_{\text{pref}}/L_{\text{all}}$, representing the proportion of the route’s total length containing the given attribute. For each preferred attribute, the highest-scoring route in the survey route’s choice set was selected as the route for that attribute. The same route could be selected for more than one attribute if it had the highest score for each attribute.

Table 3: Preferred attributes by path type

Path type	Preferred attribute
Short	Length, shortest preferred
Infrastructure	Presence of any cycling infrastructure, such as cycleways or separated lanes
Speed	Speed limit of 40 km/h or less
Flat	Slope of 2% or less
Green	Tree canopy cover of 10% or more
LTS	Level of Traffic Stress 1 or 2, out of a 4-level scale where 1 is least stressful

The remaining BFSLE and random weight perturbation routes were then ranked in order of their commonality scores against the other routes in the choice set; the 5 BFSLE and 5 RWP routes with the lowest maximum commonality scores for each survey route were retained in the choice set; and the remaining BFSLE and RWP routes were discarded. Figure 2 illustrates an example of a generated choice set for a sample origin–destination pair.

3.5 Path-size correction for overlapping routes

We estimate utility coefficients using the PSL model, each augmented with a PS correction. PSL extends the standard MNL to temper the Independence of Irrelevant Alternatives (IIA) assumption when candidate routes share portions of their paths. In route-choice modelling, alternatives often overlap on common links, which induces correlation among utilities and can violate IIA. The PS correction explicitly accounts for this by reducing the contribution of routes that substantially overlap with others. Although some overlap is inevitable in urban cycling networks, including a PS term ensures that this dependence is represented in the estimation.

To model overlap within a given choice set, we add a PS component to the deterministic utility. For route r in choice set C , let Γ_r denote the set of links composing the route, l_a the length of link $a \in \Gamma_r$, and $L_r = \sum_{a \in \Gamma_r} l_a$ the total route length. Define $\delta_{ar} = 1$ if link a belongs to route r (and 0 otherwise), and let $n_a = \sum_{j \in C} \delta_{aj}$ denote the number of routes j in the choice set that use link a . The unlogged PS measure is

$$\widetilde{\text{PS}}_r = \sum_{a \in \Gamma_r} \frac{l_a}{L_r} \cdot \frac{1}{n_a}. \quad (1)$$

This term equals 1 for a fully distinct route and $1/|C|$ when all links are shared across the choice set.

Following the standard formulation, the logged value enters the utility:

$$\text{PS}_r = \ln(\widetilde{\text{PS}}_r), \quad \text{PS}_r \in [\ln(1/|C|), 0]. \quad (2)$$

Thus, greater overlap yields a more negative PS_r . The deterministic utility for route r is:

$$V_r = \sum_m \beta_m x_{m,r} + \beta_{\text{PS}} \text{PS}_r, \quad (3)$$

where $x_{m,r}$ are the observed route attributes and β_m are their associated coefficients. When $\beta_{\text{PS}} > 0$, routes that overlap heavily with others receive lower utility relative to more distinct alternatives. In MPSL, some β_m are

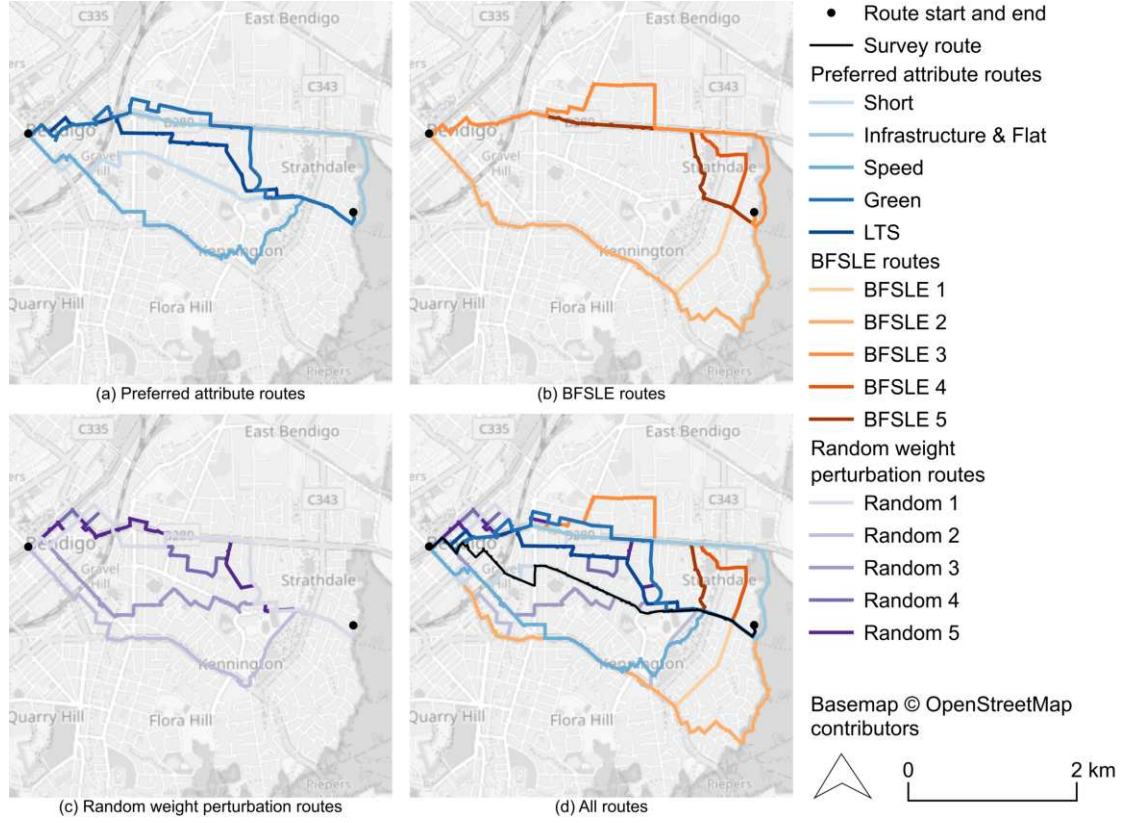


Figure 2: Example of choice set generated for a test origin and destination pair

allowed to vary across individuals to capture taste heterogeneity—that is, differences in how cyclists value particular route attributes.

3.6 Route choice model

Cyclists' route choice behaviour was analysed using two complementary model formulations: PSL and its extended MPSL. Both models estimate the probability of selecting a specific route from a set of generated feasible alternatives based on the perceived utility of each route.

The utility U_r associated with route r is expressed as:

$$U_r = \beta_{\text{can}} x_{\text{can},r} + \beta_{\text{slope}} x_{\text{slope},r} + \beta_{\text{LTS2}} x_{\text{LTS2},r} + \beta_{\text{LTS3}} x_{\text{LTS3},r} + \beta_{\text{LTS4}} x_{\text{LTS4},r} + \beta_{\text{int}} x_{\text{int},r} + \beta_{\text{det}} x_{\text{det},r} + \beta_{\text{PS}} (PS_r) + \xi_r, \quad (4)$$

where ξ_r denotes the unobserved random component of utility, assumed to follow an independent and identically distributed Gumbel distribution. The logarithmic PS term (PS_r) accounts for route overlap within each choice set, as discussed in Section 3.5.

The explanatory variables included in Equation 4, along with their definitions and measurement units, are summarised in Table 4. The associated coefficients ($\beta_{\text{can}}, \beta_{\text{slope}}, \beta_{\text{LTS2}}, \beta_{\text{LTS3}}, \beta_{\text{LTS4}}, \beta_{\text{int}}, \beta_{\text{det}}, \beta_{\text{PS}}$) are to be estimated through model calibration.

Table 4: Description of attributes.

Attribute	Description
Canopy cover ($x_{\text{can},r}$)	Weighted average canopy fraction (0–1) along the route, representing the extent of natural shade and greenery.
Maximum slope ($x_{\text{slope},r}$)	Maximum change in elevation (%) across all links forming the route.
LTS 2 ($x_{\text{LTS2},r}$)	Proportion of route length with LTS 2, corresponding to low-stress cycling environments suitable for most adults.
LTS 3 ($x_{\text{LTS3},r}$)	Proportion of route length with LTS 3, representing moderate-stress cycling conditions suitable for confident cyclists.
LTS 4 ($x_{\text{LTS4},r}$)	Proportion of route length with LTS 4, indicating high-stress or traffic-dominated cycling segments.
Intersection density ($x_{\text{int},r}$)	Density of intersections along the route, calculated as the total number of intersections divided by route length (intersections/km), representing network connectivity and accessibility.
Detour ratio ($x_{\text{det},r}$)	Ratio of the chosen route length to the shortest possible path length, reflecting the deviation or indirectness of the route.
PS (PS_r)	Degree of spatial overlap between route alternatives within the choice set, representing the distinctiveness of each route.

3.6.1 Path Size Logit (PSL) model

The PSL model assumes that the coefficients β are fixed across individuals, implying homogeneous preferences. Given the Gumbel-distributed error term, the probability that route r is selected from the choice set C is expressed as:

$$P_r = \frac{\exp(V_r)}{\sum_{k \in C} \exp(V_k)}, \quad (5)$$

where V_r denotes the deterministic component of utility defined in Equation 4. Model parameters β are estimated by maximum likelihood, with the log-likelihood function given by:

$$\ln L(\beta) = \sum_{n=1}^N \sum_{r \in C_n} \ln P_r, \quad (6)$$

where the observed choice for each cyclist is included in the summation, estimation is carried out using the *apollo* package (Hess and Palma, 2019), which implements quasi-Newton optimisation. In particular, the BFGS (Broyden–Fletcher–Goldfarb–Shanno) algorithm is employed to maximise the log-likelihood (Fletcher, 2000).

In summary, the PSL model provides a baseline representation of cyclists’ average route preferences under the assumption of homogeneous behaviour, and serves as the point of comparison for the mixed specification introduced in the following subsection.

3.6.2 Mixed Path Size Logit (MPSL) model

While the PSL model corrects for route overlap, it assumes homogeneous preferences across cyclists. To capture unobserved heterogeneity in sensitivities to route attributes, we employ the MPSL model. In this formulation, coefficients are treated as random and vary across individuals according to continuous probability distributions:

$$\beta_m = \mu_m + \sigma_m \eta_m, \quad \eta_m \sim \mathcal{N}(0, 1), \quad (7)$$

where μ_m and σ_m denote the population mean and standard deviation of coefficient m . A statistically significant σ_m indicates meaningful heterogeneity in route preferences. Conditional on a specific draw of random coefficients $\beta^{(r)}$, the choice probability retains the same logit structure as in Equation 5. The unconditional probability integrates over the distribution of random parameters:

$$P_r = \int P_{r|\beta} f(\beta | \mu, \Sigma) d\beta, \quad (8)$$

which has no closed form and is approximated via simulation. Specifically, we use quasi-random Sobol draws (Sobol, 1967) as follows:

$$\hat{P}_r = \frac{1}{R} \sum_{s=1}^R P_{r|\beta^{(s)}}, \quad (9)$$

with R denoting the number of simulation draws. In this study, 300 Sobol draws were used to balance estimation precision and computational efficiency.

Model parameters (μ_m, σ_m) are estimated by maximum simulated likelihood using the *Apollo* package in *R* (Hess and Palma, 2019). Optimisation was performed using the BFGS quasi-Newton algorithm, similar to the one in PSL. The estimated means (μ_m) represent the average influence of route attributes across cyclists, while the estimated standard deviations (σ_m) capture the degree of unobserved heterogeneity in preferences.

4 Results

4.1 Descriptive results

Before examining model estimates, we compare the characteristics of the observed routes with those of the generated alternatives to ensure that the choice sets represent realistic variation around cyclists' actual selections. Table 5 summarises these differences numerically, and Figure 3 shows the corresponding distributions.

As shown in Table 5, chosen routes are generally shorter, with a mean of 4.89 km (P95 13.0 km) compared to 4.94 km (P95 12.5 km) for alternatives, and exhibit lower detour ratios (P95 0.48 vs. 0.54). This reflects cyclists' preference for direct and time-efficient connections. Terrain differences are also evident, with chosen routes being slightly flatter on average (4.82% vs. 5.61%), suggesting that cyclists tend to avoid steep gradients when possible.

Safety and comfort considerations are evident in the LTS distribution. Chosen routes include a higher proportion of low-stress segments (LTS 2: 0.402 vs. 0.484) and fewer high-stress sections (LTS 3–4: 0.303 vs. 0.311) than alternatives, indicating a consistent avoidance of stressful cycling environments. Differences also emerge in terms of network connectivity. Chosen routes have slightly lower average intersection densities than the generated alternatives (6.71 vs. 7.04). PS values are marginally higher for observed routes (0.339 vs. 0.326), indicating that chosen routes tend to be more distinctive and less overlapping with other alternatives. In contrast, canopy cover is slightly higher

for the observed than for the alternative routes (0.202 vs. 0.194), suggesting a preference for greener routes, although the difference is smaller than for some other attributes.

Figure 3 visually compares the distributions of key route attributes for observed and alternative routes. Across all attributes, the overall shapes and central tendencies are broadly aligned, indicating that the alternatives span a similar range of values to the routes cyclists actually selected. At the same time, the alternatives generally exhibit greater spread—particularly for distance, detour ratio, slope, and LTS attributes, which reflect a wider range of plausible routing conditions. Notably, the distributions for LTS 2 show the visual separation between observed and alternative routes, with high-stress segments (LTS 3–4) also exhibiting distinct patterns. This separation is consistent with the strong statistical significance of LTS variables in the route choice models. Blue bars represent observed routes, orange bars denote alternatives, and darker blended bars indicate overlapping bins.

Table 5: Descriptive statistics of route attributes for surveyed and alternative routes.

Attribute	Surveyed				Alternatives in the choice set			
	Mean	Median	P95	Max	Mean	Median	P95	Max
Distance (km)	4.890	3.570	13.000	25.300	4.940	3.750	12.500	26.400
Canopy cover (Fraction)	0.202	0.174	0.408	0.737	0.194	0.173	0.378	0.737
Maximum slope (%)	4.820	4.440	8.120	18.900	5.610	4.990	10.600	22.200
LTS 2 (proportion)	0.402	0.381	0.854	1.000	0.484	0.479	0.872	1.000
LTS 3 (proportion)	0.148	0.075	0.511	0.909	0.138	0.098	0.440	1.000
LTS 4 (proportion)	0.155	0.084	0.481	0.869	0.173	0.134	0.504	0.967
Intersection density	6.710	6.820	10.400	14.300	7.040	7.180	10.600	16.400
Detour ratio	0.138	0.076	0.483	1.520	0.174	0.128	0.544	1.520
PS	0.339	0.291	0.693	0.920	0.326	0.302	0.580	0.958

4.2 Multicollinearity and correlation analysis

Before estimating the route-choice models, we examine correlations among the explanatory variables to assess potential multicollinearity and to ensure that each attribute captures a distinct aspect of the cycling environment. High correlations or inflated VIFs could compromise the stability and interpretability of coefficient estimates. To this end, Pearson correlation coefficients and VIFs were computed across all route alternatives. Table 6 presents the correlation matrix, and Table 7 summarises the corresponding VIF values.

The correlation results indicate generally moderate relationships, with all absolute correlation coefficients ≤ 0.62 . Intersections density shows the strongest associations, being negatively correlated with canopy cover ($r = -0.617$) and slope ($r = -0.121$), while positively correlated with LTS 2 ($r = 0.426$), LTS 4 ($r = 0.133$), and PS ($r = 0.097^{***}$). These relationships reflect the structure of denser, well-connected networks: routes with more intersections per kilometre typically occur in fine-grained urban grids that enhance connectivity and expand route choice sets (higher PS values), though at the cost of slightly greater exposure to traffic stress.

Other attributes are moderately correlated with one another. Canopy cover is positively related to slope ($r = 0.162^{***}$), suggesting that greener corridors often coincide with hillier terrain, while both are negatively correlated with low-stress cycling segments (LTS 2). The LTS variables exhibit the expected inverse relationships, where routes

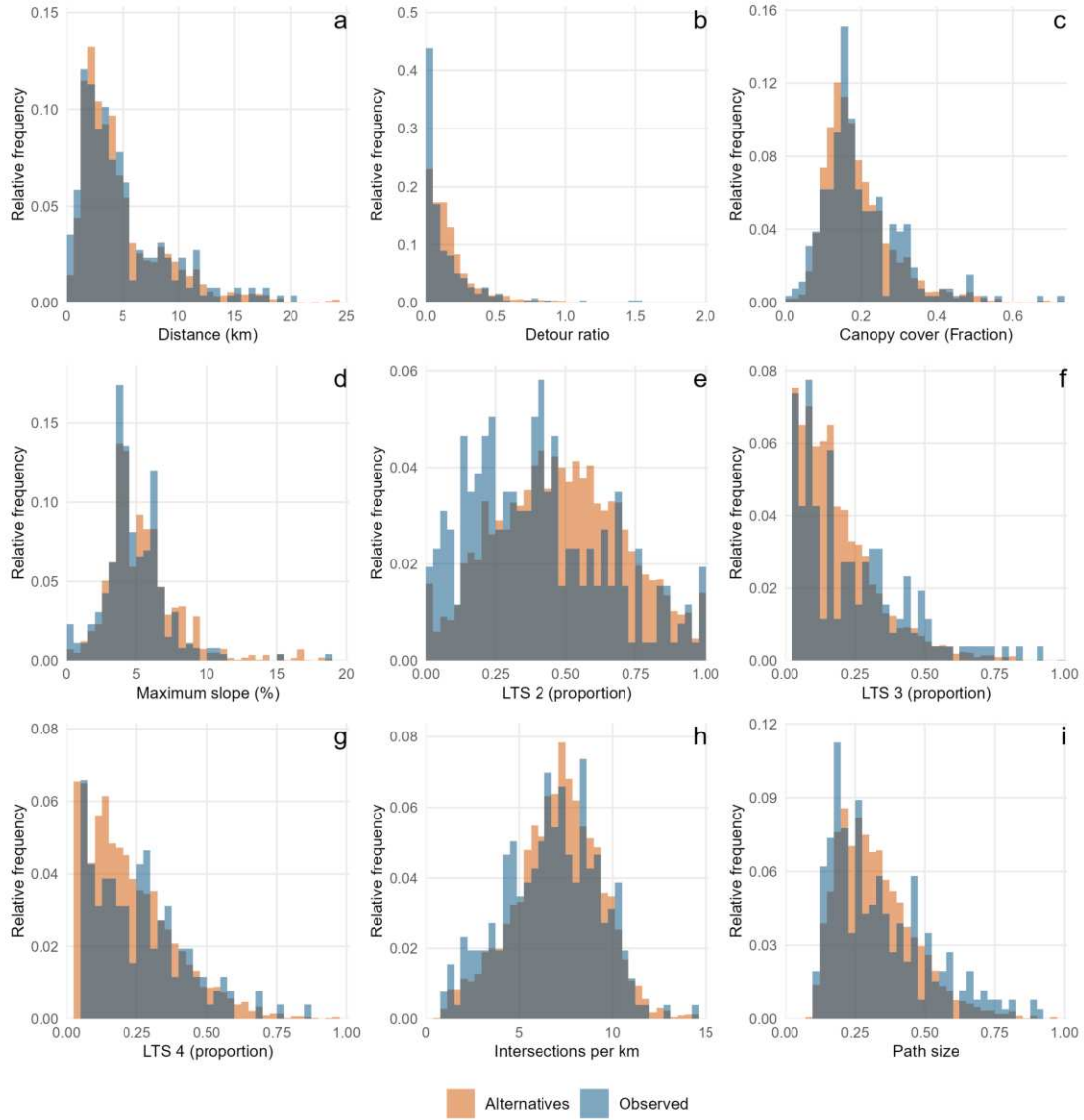


Figure 3: Distribution of route attributes for observed (chosen) and generated alternative routes: (a) Distance (km), (b) Detour ratio, (c) Canopy cover (%), (d) Maximum slope (%), (e–g) LTS 2–4 (proportion), (h) Number of intersections, and (i) PS. Blue bars represent chosen routes and orange bars alternatives; overlapping bins appear in darker shades.

with higher proportions of low-stress links (LTS 2) tend to have fewer moderate and high-stress sections (LTS 3–4), consistent with distinct roadway typologies and traffic stress patterns across the network.

Overall, these findings confirm that the explanatory variables capture related but distinct aspects of the cycling environment, with no indication of redundancy. The VIF results in Table 7 further support this conclusion: all values

remain comfortably below standard multicollinearity thresholds, with the highest being 2.57 for LTS 2, followed by canopy cover (1.92), which exhibits remarkably low VIFs. This indicates that each attribute contributes largely independent information to explaining cyclists' route-choice behaviour.

Table 6: Pearson correlation matrix of route attributes.

	Canopy	Maximum slope	LTS 2	LTS 3	LTS 4	Intersections density	Detour	PS
Canopy	1.000***	0.162***	-0.332***	-0.089***	-0.202***	-0.617***	0.043***	-0.067***
Maximum slope	0.162***	1.000***	-0.121***	0.102***	-0.113***	-0.121***	-0.037**	0.097***
LTS 2	-0.332***	-0.121***	1.000***	-0.372***	-0.409***	0.426***	0.076***	0.142***
LTS 3	-0.089***	0.102***	-0.372***	1.000***	-0.050***	-0.038**	-0.074***	0.013
LTS 4	-0.202***	-0.113***	-0.409***	-0.050***	1.000***	0.133***	-0.057***	0.002
Intersections density	-0.617***	-0.121***	0.426***	-0.038**	0.133***	1.000***	0.031**	0.097***
Detour	0.043***	-0.037**	0.076***	-0.074***	-0.057***	0.031**	1.000***	0.427***
PS	-0.067***	0.097***	0.142***	0.013	0.002	0.097***	0.427***	1.000***

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 7: Variance Inflation Factor (VIF) results.

Attribute	VIF
LTS 2	2.565***
Canopy	1.919***
Intersections per km	1.866***
LTS 4	1.791***
LTS 3	1.475***
PS	1.304***
Detour	1.260***
Maximum slope	1.085***

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

4.3 Route choice model estimation results

4.3.1 PSL model results

Table 8 presents the estimation results of both PSL and MPSL models. The PSL model shows a satisfactory fit for a route choice model, with a R^2 of 0.1764. This indicates that the included route-level attributes capture key aspects of cyclists' observed route choices under the PSL model.

Most estimated attributes follow expected behaviour. Attributes related to traffic stress (LTS 2, LTS 3, and LTS 4) are all negative and highly significant, confirming that cyclists are less likely to select routes with higher levels of traffic stress exposure. Among these, LTS 2 has the strongest negative value ($\beta_{\text{LTS2}} = -1.800$), indicating a strong aversion to even moderately stressful segments relative to the least stressful LTS 1 category, while LTS 4 ($\beta_{\text{LTS4}} = -1.232$ and LTS 3 ($\beta_{\text{LTS3}} = -0.707$) also exert powerful adverse effects. These findings confirm that

cyclists strongly prefer calm and protected environments, avoiding busy corridors or locations with high vehicular interaction, and that in regional cities, even LTS 2 conditions substantially reduce route utility.

The maximum slope variable is negative and significant ($\beta_{\text{slope}} = -0.745$), indicating that steeper gradients discourage route selection due to the additional physical effort required. The narrow confidence interval suggests a consistent effect across users. In contrast, the intersection density variable is positive and significant ($\beta_{\text{int}} = 0.453$), implying that routes with more intersections per km are more attractive, likely because they occur within finer-grained regional networks that offer higher connectivity and accessibility, but also lower relative speeds of motor vehicles.

The canopy cover coefficient ($\beta_{\text{can}} = -0.411$) is statistically significant and negative, which is counterintuitive, suggesting that shaded routes in this regional context may be associated with lower network directness rather than comfort alone. The detour ratio remains negative and highly significant ($\beta_{\text{det}} = -0.860$), indicating a strong preference for direct and time-efficient routes. This pattern suggests that cyclists are willing to accept moderate exposure to traffic stress or slope when doing so reduces overall travel distance or travel time. Finally, the PS coefficient is positive and highly significant ($\beta_{\text{ps}} = 0.567$), indicating sensitivity to route overlap within the choice set, capturing a behavioural tendency to discount routes that substantially share the same links as others. This finding is consistent with the intended role of the PS term in accounting for correlation among overlapping alternatives in PSL models.

Overall, the PSL results showed that cyclists in CoGB prioritise low-stress, direct, and well-connected routes while avoiding steep and indirect paths.

4.3.2 MPSL model results

The MPSL model extends the PSL model by allowing all route attributes to vary randomly across individuals, thereby capturing unobserved heterogeneity in cyclists' route preferences. The model provides an improved fit relative to the PSL, with a higher R^2 of 0.2908, confirming substantial behavioural variation among cyclists and highlighting the value of accounting for preference heterogeneity in improving explanatory power and behavioural realism.

The mean coefficients (μ_{β}) of the MPSL model follow the same trends as the PSL results but with higher magnitudes. All three LTS attributes remain negative and highly significant, reaffirming that cyclists consistently avoid stressful environments. The larger magnitudes for LTS 2 ($\mu_{\text{LTS2}} = -17.34$) and LTS 4 ($\mu_{\text{LTS4}} = -18.328$) indicate stronger aversion to high-stress conditions once individual variability is considered. Similarly, the slope parameter remains negative and significant ($\mu_{\text{slope}} = -0.553$), reinforcing that the gradient is a key deterrent for route choice. The PS coefficient ($\mu_{\text{ps}} = 2.822$) increases relative to the PSL, highlighting that cyclists distinctly value routes with less spatial overlap and greater individuality. The detour ratio shows the most pronounced effect ($\mu_{\text{det}} = -21.75$), suggesting that while some riders are strongly averse to indirect routes, others may willingly accept longer paths when compensated by comfort or safety. The coefficient for intersection density ($\mu_{\text{int}} = 0.335$) is positive but not statistically significant, indicating a modest yet diverse influence of network connectivity on route attractiveness.

The estimated standard deviations (σ) of the random parameters demonstrate substantial heterogeneity in preferences. Several parameters, including σ_{det} , σ_{can} , σ_{int} , σ_{LTS2} , and σ_{LTS4} are significant, revealing large variability among cyclists in their sensitivities to these factors. The significant standard deviation for σ_{can} (-28.571) suggests divergent perceptions of canopy cover: while some riders may value shaded and visually pleasant routes, others appear indifferent, prioritising efficiency instead. Likewise, the large and highly significant spread in σ_{det} (31.855) indicates diverse tolerance levels toward indirect routes. The positive and significant σ_{LTS2} , σ_{LTS4} (6.914, and 9.006) reflect strong heterogeneity in sensitivity to high-stress links, suggesting that some cyclists entirely avoid them, while others tolerate limited exposure when necessary.

Table 8: Comparison of PSL and MPSL models.

Attribute	PSL			MPSL		
	Estimate	t-stat	95% CI	Estimate	t-stat	95% CI
Fixed coefficients in PSL (β_r) and mean coefficients in MPSL (μ_r)						
$\beta_{\text{can}} \mid \mu_{\text{can}}$	-0.411	-2.369**	[-0.753, -0.071]	-5.379	-1.182	[-14.300, 3.542]
$\beta_{\text{slope}} \mid \mu_{\text{slope}}$	-0.745	-4.533***	[-1.067, -0.423]	-0.553	-3.762***	[-0.841, -0.265]
$\beta_{\text{LTS2}} \mid \mu_{\text{LTS2}}$	-1.800	-9.957***	[-2.155, -1.446]	-17.340	-4.235***	[-25.366, -9.315]
$\beta_{\text{LTS3}} \mid \mu_{\text{LTS3}}$	-0.707	-5.547***	[-0.957, -0.457]	-11.518	-3.635***	[-17.730, -5.307]
$\beta_{\text{LTS4}} \mid \mu_{\text{LTS4}}$	-1.232	-8.385***	[-1.520, -0.944]	-18.328	-4.174***	[-26.935, -9.721]
$\beta_{\text{int}} \mid \mu_{\text{int}}$	0.453	2.184**	[0.046, 0.859]	0.335	1.458	[-0.115, 0.785]
$\beta_{\text{det}} \mid \mu_{\text{det}}$	-0.860	-3.969***	[-1.285, -0.435]	-21.750	-3.846***	[-32.834, -10.666]
$\beta_{\text{PS}} \mid \mu_{\text{PS}}$	0.567	6.812***	[0.404, 0.731]	2.822	4.838***	[1.679, 3.965]
MPSL standard deviations of random parameters (σ_r)						
σ_{can}		—		-28.571	-3.201***	[-46.064, -11.078]
σ_{slope}		—		0.373	2.507**	[0.081, 0.664]
σ_{LTS2}		—		6.914	1.953*	[-0.026, 13.854]
σ_{LTS3}		—		8.045	1.625	[-1.660, 17.749]
σ_{LTS4}		—		9.006	2.730***	[2.539, 15.472]
σ_{int}		—		-0.896	-1.897*	[-1.822, 0.030]
σ_{det}		—		31.855	4.141***	[16.778, 46.932]
σ_{PS}		—		0.116	0.779	[-0.175, 0.406]
Model fit						
R^2		0.1764			0.2908	
LL		-597.06			-512.73	
AIC		1210.13			1057.45	

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. PSL coefficients are denoted by β_r ; MPSL mean coefficients by μ_r , with standard deviations σ_r .

Variability in intersection density ($\sigma_{\text{int}} = -0.896$) is marginally significant, implying differences in how cyclists perceive intersections. Some associate them with improved connectivity, whereas others view them as interruptions or potential points of conflict. The slope parameter ($\sigma_{\text{slope}} = 0.373$) is also significant, indicating that individuals exhibit varying levels of sensitivity to gradients. In contrast, the PS term ($\sigma_{\text{PS}} = 0.116$, $t = 0.779$) remains insignificant, indicating relatively limited heterogeneity in how cyclists account for route overlap across alternatives.

Overall, the MPSL results underscore the importance of recognising individual differences in route choice behaviour. Cyclists vary widely in how they balance directness, comfort, and safety, with some strongly avoid stress and detours, while others demonstrate flexibility depending on environmental and network conditions. Accounting for this heterogeneity enhances both the explanatory and predictive capabilities of the model, thereby offering a more realistic representation of cycling route-choice behaviour.

4.4 Group-specific route choice models

In this section, we examine variations in cyclists' route choice behaviour by estimating separate PSL models for different subgroups, classified according to gender, age category, and trip purpose.

4.4.1 Gender-specific PSL models

Table 9 shows the results of the gender-wise PSL models. The signs of the coefficients for both male and female cyclists are consistent with the main PSL model: steep slopes, high traffic stress, and detours reduce route utility, whereas PS increases it. However, systematic differences in the magnitude and significance of these attributes highlight evident gender-based heterogeneity in route choice behaviour.

The coefficients for $\beta_{LTS3} = -8.215$ and $\beta_{LTS4} = -9.551$ among females are considerably larger than those for males ($\beta_{LTS3} = -2.524$ and $\beta_{LTS4} = -5.946$, respectively), indicating that female cyclists are more sensitive to traffic stress and more likely to avoid routes with high vehicular exposure or limited protective infrastructure. The slope attribute is negative and significant for both genders, with a slightly larger effect for females ($\beta_{slope} = -0.254$) than for males ($\beta_{slope} = -0.216$), indicating that physical exertion and topography act as greater deterrents for female cyclists.

Canopy cover (β_{can}) is negative but insignificant for males ($\beta_{can} = -1.020$), whereas it is negative and highly significant for females ($\beta_{can} = -7.563$). This suggests that greenery or shaded environments, often associated with local or less direct streets, may deter some female cyclists seeking efficiency or safety, reflecting a complex interaction between comfort and practicality in route preferences.

The detour ratio is negative and significant for males ($\beta_{det} = -5.815$) but statistically insignificant for females ($\beta_{det} = -2.448$). This contrast suggests that male cyclists tend to prioritise route efficiency and directness, whereas female cyclists may be more willing to accept longer routes if they offer perceived safety or comfort benefits. The PS coefficient is positive and highly significant for both genders, with a slightly higher value for females ($\beta_{ps} = 1.197$) compared to males ($\beta_{ps} = 1.154$). This indicates that both male and female cyclists tend to prefer routes that are distinctive and less overlapping with other alternatives.

The intersection density coefficient (β_{int}) is positive and significant for females ($\beta_{int} = 0.462$, $t = 2.821^{***}$) but insignificant for males ($\beta_{int} = 0.033$). This suggests that, after controlling for other attributes, female cyclists have an underlying preference for routes with higher intersection density, likely reflecting greater utility for connectivity and accessibility in finer-grained urban networks. In contrast, the lack of significance among males suggests that their route preferences are primarily determined by measurable physical attributes such as detour and traffic stress.

In summary, while both genders share the general preference for safe, direct, and distinctive routes, the strength and significance of these preferences differ. Female cyclists are significantly more risk-averse, particularly regarding traffic stress and slope, whereas male cyclists exhibit greater tolerance but place relatively greater value on directness and efficiency. These results highlight gender-based heterogeneity in route choice behaviour and underscore the importance of designing cycling infrastructure that balances the needs of both safety-sensitive and efficiency-oriented users.

4.4.2 Age group specific PSL models

The results presented in Table 10 reveal meaningful behavioural differences in route choice preferences across the three age groups. Overall, all groups exhibit the same fundamental behavioural patterns as the main PSL model, steep slopes ($\beta_{slope} < 0$), detours ($\beta_{det} < 0$), and traffic stress ($\beta_{LTS2-4} < 0$) reduce utility, while PS ($\beta_{ps} > 0$) enhances

Table 9: Estimation results of gender-specific PSL models.

Attribute	Male			Female		
	Estimate	t-stat	95% CI	Estimate	t-stat	95% CI
β_{can}	-1.020	-0.407	[-5.938, 3.897]	-7.563	-2.786***	[-12.884, -2.242]
β_{slope}	-0.216	-3.686***	[-0.331, -0.101]	-0.254	-3.130***	[-0.413, -0.095]
β_{LTS2}	-6.681	-6.355***	[-8.741, -4.620]	-10.178	-7.118***	[-12.980, -7.375]
β_{LTS3}	-2.524	-2.467**	[-4.530, -0.519]	-8.215	-5.724***	[-11.027, -5.402]
β_{LTS4}	-5.946	-5.121***	[-8.221, -3.670]	-9.551	-6.646***	[-12.367, -6.734]
β_{int}	0.033	0.301	[-0.182, 0.248]	0.462	2.821***	[0.141, 0.783]
β_{det}	-5.815	-5.177***	[-8.016, -3.613]	-2.448	-1.083	[-6.876, 1.980]
β_{PS}	1.154	4.328***	[0.631, 1.676]	1.197	3.117***	[0.444, 1.950]
Model fit:						
R^2		0.1724			0.1889	
LL		-344.69			-246.28	
AIC		705.39			508.55	

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

it. However, the magnitudes differ between different age groups, highlighting systematic heterogeneity across age categories. Model fit reinforces these differences: the youngest group (18–40 years) achieves the highest explanatory power ($R^2 = 0.2225$), followed by middle-aged riders (40–60 years, $R^2 = 0.1830$), while the oldest group (> 60 years) yields the weakest fit ($R^2 = 0.1055$). This suggests that measurable route attributes best explain younger cyclists' choices, whereas unobserved or contextual factors may play a larger role among older riders.

Middle-aged cyclists (40–60 years) exhibit a very strong aversion to traffic stress, with significant negative coefficients for LTS 2 ($\beta_{\text{LTS2}} = -8.954$) and LTS 4 ($\beta_{\text{LTS4}} = -8.006$), indicating a clear preference for protected routes. Younger cyclists (18–40 years) show a similarly strong pattern, with slightly smaller aversion to LTS 2 ($\beta_{\text{LTS2}} = -8.187$) but a marginally larger aversion to LTS 4 ($\beta_{\text{LTS4}} = -8.169$). Overall, both age groups are highly sensitive to stressful traffic environments. The oldest cyclists (> 60 years) also avoid high-stress routes, though with weaker magnitudes ($\beta_{\text{LTS2}} = -4.796$, $\beta_{\text{LTS4}} = -4.216$), implying that while safety remains important, their preferences may be moderated by familiarity, comfort, or limited exposure to diverse route options.

The canopy cover variable displays an interesting age-related pattern. It is negative and insignificant among younger riders ($\beta_{\text{can}} = -2.052$), suggesting a limited influence of greenery on utilitarian route choice. Among middle-aged cyclists, canopy cover becomes a significant negative factor ($\beta_{\text{can}} = -5.471$), suggesting that shaded routes in CoGB may correspond to slower, less direct corridors. For the oldest group, the effect remains negative and insignificant ($\beta_{\text{can}} = -1.980$), suggesting that environmental comfort plays a minimal role once key structural attributes are taken into account.

The slope variable is negative and significant across all age groups, with the most substantial effect among younger cyclists ($\beta_{\text{slope}} = -0.292$), followed by middle-aged ($\beta_{\text{slope}} = -0.243$) and older riders ($\beta_{\text{slope}} = -0.180$). This trend suggests that younger cyclists may encounter more varied terrain due to longer or more diverse travel patterns, whereas older riders may restrict their travel to familiar, flatter areas.

The detour coefficient is negative across all groups, indicating a shared preference for shorter, more direct routes.

However, its statistical significance differs across age groups: the effect is substantial and statistically significant for middle-aged ($\beta_{\text{det}} = -5.320$) and older riders ($\beta_{\text{det}} = -3.218$), but less statistically significant for younger cyclists ($\beta_{\text{det}} = -4.592$). This implies that middle-aged cyclists (40–60 years) place the strongest emphasis on route efficiency, while younger riders may be more flexible in trading off distance for comfort or variety. Intersection density is insignificant for younger riders ($\beta_{\text{int}} = -0.086$), but positive and significant for both middle-aged ($\beta_{\text{int}} = 0.294$) and older cyclists ($\beta_{\text{int}} = 0.309$), suggesting that these groups value connectivity and accessibility in finer-grained street networks. This may also reflect risk mitigation through familiar, well-connected areas that facilitate safer navigation and shorter deviations.

The PS coefficient is positive and significant across all age groups ($\beta_{\text{PS}} = 1.098, 1.331^*, \text{ and } 0.917^{**}$, respectively), indicating that more distinctive and less overlapping routes are consistently preferred. The slightly higher magnitude among middle-aged cyclists highlights their greater sensitivity to route uniqueness, possibly reflecting awareness of available network options or experience-based route refinement.

In summary, age-wise models highlight evident heterogeneity in route-choice behaviour. Younger cyclists appear somewhat more flexible and adaptive, with weaker responses to traffic stress and environmental attributes. Middle-aged cyclists are the most infrastructure-conscious, exhibiting a strong aversion to stress and detours and prioritising direct, efficient routes. Older cyclists remain safety-sensitive but display behaviour that may be influenced by routine and comfort. These insights underscore the importance of age-sensitive transport planning that balances the needs of younger cyclists seeking flexibility with the safety and efficiency requirements of middle-aged and older users.

Table 10: Estimation results of age-specific PSL models.

Attribute	18–40 years			40–60 years			>60 years		
	Estimate	t-stat	95% CI	Estimate	t-stat	95% CI	Estimate	t-stat	95% CI
β_{can}	-2.052	-0.529	[-9.665, 5.560]	-5.471	-2.228**	[-10.284, -0.658]	-1.980	-0.501	[-9.730, 5.770]
β_{slope}	-0.292	-3.093***	[-0.477, -0.107]	-0.243	-2.952***	[-0.404, -0.082]	-0.180	-2.311**	[-0.332, -0.027]
β_{LTS2}	-8.187	-5.903***	[-10.905, -5.469]	-8.954	-7.802***	[-11.203, -6.704]	-4.796	-2.867***	[-8.074, -1.518]
β_{LTS3}	-4.464	-2.590***	[-7.841, -1.086]	-4.548	-3.495***	[-7.099, -1.997]	-4.258	-2.790***	[-7.249, -1.266]
β_{LTS4}	-8.169	-4.880***	[-11.450, -4.888]	-8.006	-6.331***	[-10.484, -5.527]	-4.216	-2.170**	[-8.024, -0.408]
β_{int}	-0.086	-0.477	[-0.441, 0.268]	0.294	2.147**	[0.026, 0.563]	0.309	1.993**	[0.005, 0.613]
β_{det}	-4.592	-1.643*	[-10.071, 0.887]	-5.320	-2.413**	[-9.640, -0.999]	-3.218	-2.258**	[-6.012, -0.424]
β_{PS}	1.098	2.281**	[0.155, 2.041]	1.331	3.377***	[0.559, 2.104]	0.917	2.502**	[0.199, 1.636]
Model fit:									
R^2		0.2225			0.1830			0.1055	
LL		-173.14			-287.39			-130.30	
AIC		362.27			590.78			276.59	

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

4.4.3 Trip purpose specific PSL models

The results in Table 11 reveal clear behavioural contrasts between commute and non-commute trips, where commute refers to work-related travel and non-commute captures other utilitarian purposes. Both groups follow the same general tendencies as the main PSL model, steep slopes ($\beta_{\text{slope}} < 0$), detours ($\beta_{\text{det}} < 0$), and traffic stress ($\beta_{\text{LTS2-4}} < 0$) reduce utility, while PS ($\beta_{\text{PS}} > 0$) enhances it. However, the magnitude of responses differs sharply, reflecting the distinct priorities of each trip purpose. The model fit underscores this difference: the commute model achieves substantially higher explanatory power ($R^2 = 0.243$) than the non-commute model ($R^2 = 0.1268$), suggesting that the choices of cyclists who commute are more systematically influenced by measurable route attributes.

Commuters are particularly risk-averse, with substantial and significant coefficients for traffic stress: $\beta_{\text{LTS2}} = -11.672$ and $\beta_{\text{LTS4}} = -10.690$, compared to $\beta_{\text{LTS2}} = -5.874$ and $\beta_{\text{LTS4}} = -5.315$ for non-commuters. This suggests that stressful environments are particularly prohibitive for work-related trips, where predictability and safety are crucial during travel. Non-commuters display greater tolerance for mixed-traffic conditions.

The detour coefficient is significantly higher for commuters ($\beta_{\text{det}} = -5.931$) than for non-commuters ($\beta_{\text{det}} = -3.635$), highlighting the focus on directness and time minimisation during travel. This contrasts with non-commute trips, where longer or more scenic routes may be acceptable, particularly when time constraints are less binding.

Slopes reduce utility across both models and are statistically significant ($\beta_{\text{slope}} = -0.257$ for commuters; $\beta_{\text{slope}} = -0.207$ for non-commuters), underscoring that elevation changes deter cyclists irrespective of trip purpose. The similar magnitudes suggest that topography imposes a consistent physical constraint across trip types, though commuters' stronger t-statistic reflects slightly greater sensitivity to steep gradients.

Canopy cover is negative in both models, remaining statistically insignificant for commute trips ($\beta_{\text{can}} = -3.853$) but significant for non-commute trips ($\beta_{\text{can}} = -5.504^{**}$). This suggests that shaded or scenic routes often coincide with slower or more indirect segments, reducing their attractiveness to efficiency-oriented commuters and even diminishing their appeal to leisure riders when these routes compromise directness.

PS coefficients are positive and highly significant across both models ($\beta_{\text{PS}} = 1.041$ for commuters; $\beta_{\text{PS}} = 1.272$ for non-commuters), confirming that distinctive, less-overlapping routes are consistently favoured. The slightly higher value for non-commuters suggests greater responsiveness to network variety and to exploration outside habitual or time-constrained travel.

Intersection density exhibits divergent effects: it is positive and significant for commuters ($\beta_{\text{int}} = 0.492$) but insignificant for non-commuters ($\beta_{\text{int}} = -0.025$). This implies that connectivity and route-choice flexibility are particularly valued when travelling to work. In contrast, for discretionary travel, intersections may serve as interruptions or potential conflict points rather than as accessibility advantages.

In summary, commuters are highly sensitive to traffic stress, slopes, and detours, underscoring their preference for safe, direct, and efficient routes. Non-commuters share these general tendencies but exhibit more relaxed thresholds, with greater emphasis on route uniqueness and moderate tolerance for traffic exposure. These findings highlight trip purpose as a key source of behavioural heterogeneity in cycling route choice, reinforcing the need for differentiated planning. Commuting networks should prioritise direct, high-comfort corridors, while non-commute cycling infrastructure may benefit from more scenic, flexible, and diverse route options.

5 Discussion

This study confirms that safety, comfort, and network continuity are central determinants of cyclists' route choices in a regional setting. Consistent with findings from larger metropolitan contexts (Broach et al., 2012; Prato et al., 2018; Łukawska, 2024), cyclists in the CoGB strongly avoid stressful traffic conditions, steep gradients, and indirect routes. Some of the slope-related effects may also reflect the presence of e-bike users in the sample, who typically exhibit lower sensitivity to gradients, consistent with previous findings (Lilasathapornkit et al., 2025). Under both the PSL and MPSL models, links with higher LTS levels, greater maximum slope, and larger detour ratios were systematically associated with lower route utility. In contrast, paths that were more distinctive relative to alternatives (higher PS) were preferred. The improved fit of the MPSL relative to the PSL model underscores that allowing for preference heterogeneity substantially increases behavioural realism, especially in a heterogeneous cycling population with varied abilities, risk perceptions, and trip purposes.

Table 11: Estimation results of commute and non-commute trip PSL models.

Attribute	Commute trips			Non-commute trips		
	Estimate	t-stat	95% CI	Estimate	t-stat	95% CI
β_{can}	-3.853	-1.224	[-10.022, 2.317]	-5.504	-2.376**	[-10.046, -0.963]
β_{slope}	-0.257	-2.914***	[-0.430, -0.084]	-0.207	-3.205***	[-0.333, -0.080]
β_{LTS2}	-11.672	-8.612***	[-14.329, -9.015]	-5.874	-6.204***	[-7.730, -4.018]
β_{LTS3}	-6.065	-4.540***	[-8.684, -3.447]	-3.906	-3.469***	[-6.113, -1.699]
β_{LTS4}	-10.690	-7.908***	[-13.340, -8.041]	-5.315	-4.739***	[-7.513, -3.117]
β_{int}	0.492	3.362***	[0.205, 0.778]	-0.025	-0.217	[-0.249, 0.200]
β_{det}	-5.931	-2.401**	[-10.773, -1.090]	-3.635	-2.598***	[-6.378, -0.893]
β_{PS}	1.041	2.884***	[0.334, 1.749]	1.272	4.121***	[0.667, 1.878]
Model fit:						
R^2		0.2441			0.1268	
LL		-252.83			-339.10	
AIC		521.66			694.20	

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The results further suggest that cyclists assess route alternatives in relative terms, weighing directness against exposure to stress rather than considering these attributes in isolation. While low-stress conditions are generally preferred, previous work indicates that cyclists may accept higher traffic stress when direct routes offer clear efficiency benefits or perceived legibility, such as the presence of marked facilities on shared roads (Broach et al., 2012; Łukawska, 2024). In contrast, low-stress detours appear less attractive when they do not clearly distinguish themselves from more direct options in overall exposure. In a regional setting such as the CoGB, this trade-off is likely shaped by the underlying network structure: lower road and link densities mean that alternative routes often require substantially longer deviations than in dense metropolitan networks. As a result, the strong aversion to detours observed in this study may reflect the limited availability of reasonably direct low-stress alternatives rather than a simple preference for shortest paths.

Similarly, the positive association between intersection density and route choice suggests that intersections in regional cities play a positive role by improving connectivity (Łukawska et al., 2023; Łukawska, 2024). Frequent intersections shorten road segment lengths and increase the number of feasible paths, thereby reducing detours and enabling more flexible routing. Although intersections are often perceived as high-stress environments for cyclists, higher intersection densities are also characteristic of finer-grained street networks that support lower relative vehicle speeds and greater route choice flexibility Łukawska (2024). Such networks are more common in dense, central urban areas, while more car-dependent regional cities, such as those in Australia, often suffer from low connectivity (Giles-Corti et al., 2022). This reinforces the importance of planning and designing for well-connected cycling networks in regional cities.

Two context-specific findings provide additional insights for regional planning. The negative association between canopy cover and route utility is likely to reflect where the canopy is located within the CoGB network rather than an inherent dislike of trees. In CoGB, high canopy cover often occurs along peri-urban green corridors, creek lines, and local suburban streets that are attractive for recreational riding but relatively indirect or poorly connected to key

destinations such as workplaces, schools, and shops. For the route choices explored in this study, these links are most likely not efficient by route, so a higher canopy is correlated with lower adequate accessibility. The result should therefore not be interpreted as a preference against greenery, but as evidence that, in a regional context, for transport-oriented trips, tree canopy is not a primary lever in shaping cyclists' route choices. Therefore, to ensure that cyclists receive the significant benefits of exposure to green space and tree canopy cover (Eisenman et al., 2021; Ren et al., 2023), planners should provide suitable canopy along cycling corridors and popular bike routes.

The group-specific models reveal substantial heterogeneity in route-choice behaviour by gender, age, and trip purpose. Gender-specific models indicate that female cyclists are more risk-averse, with larger negative coefficients on high LTS links and greater sensitivity to slope. In contrast, male cyclists place greater emphasis on directness, with stronger detour aversion and weaker responses to traffic stress, consistent with international evidence that women are more sensitive to perceived risk and value predictable, low stress environments (Lilasathapornkit et al., 2025; Meister et al., 2023; Hardinghaus and Weschke, 2022). Age specific models indicate that middle aged riders (40–60 years) show the strongest aversion to stressful traffic conditions and detours and a clear preference for denser intersection patterns, younger cyclists (18–40 years) are somewhat more willing to trade comfort for directness, and older cyclists (over 60 years) remain averse to high stress links, but with smaller coefficient magnitudes and weaker model fit. Trip purpose models further show that commuters are the most infrastructure-sensitive group, exhibiting a strong aversion to high LTS links and detours and a marked preference for intersection-dense networks. Non-commute trips, however, follow the same pattern, but with a weaker aversion to stress and detours and a slightly stronger preference for route distinctiveness. The MPSL model reinforces this picture, with significant standard deviations for detour, canopy, intersection density, and LTS parameters indicating wide variation in how cyclists balance directness, safety, and comfort, while the insignificant spread in PS suggests relatively shared preferences for route distinctiveness and a general tendency to prefer options that do not fully overlap with alternatives.

These differences suggest that network designs that are acceptable to confident, efficiency-oriented riders may still be perceived as too risky by more cautious groups. Given that a significant cohort of current bike riders are middle-aged men in countries with a low mode share for cycling, such as Australia, future bike infrastructure must plan not only seeks to match current riders' demand but also proactively creates a supportive environment for all existing and potential bike riders (Pearson et al., 2023). Although designing bike infrastructure for all ages and abilities is increasingly referenced in transport plans and strategies (Laberee et al., 2023), there remains a lack of clarity and coherence in design and implementation.

One of the main barriers to cycling uptake, particularly among less confident riders, is infrastructure discontinuity and gaps in bike network connectivity (Lowry and Loh, 2017). Jafari et al. (2025) showed that reducing speed limits on residential streets to 30 km/h can be a low-cost intervention that significantly lowers traffic stress for bike riders without significant impacts on car travel times. A suite of interventions, such as building physically separated facilities on high traffic and major roads accompanied by reduced speed limits on local streets, is an effective way to deliver continuous low traffic stress corridors that, when accompanied by appropriate education programs and regulations, can result in meaningful mode shift in cities with currently low levels of bike riding (Pearson et al., 2025).

Beyond behavioural findings, this study demonstrates the value of map-based PPGIS as a data collection method for cycling route choice in regional cities. Previous work has shown that PPGIS can support bicycle infrastructure planning and prioritisation (Lock and Pettit, 2022). Consistent with Alattar et al. (2021), we found that PPGIS is a practical tool for collecting route-level data to study cyclists' route choices and produce results comparable to those from studies based on large-scale GPS data. While such data lacks temporal information, such as travel times, it represents a cyclist's amalgamated view of a route of choice formed from multiple riding experiences,

and additionally affords rich information on perceived route quality. Furthermore, this method enables the use of a low-cost approach to capture rich route information and to explore questions about exposure to built and natural environment features and factors influencing route choice. The approach is particularly useful for smaller research groups with limited funding, as well as local government and advocacy organisations, as a pragmatic and scalable way to obtain route-level data that can be linked to infrastructure attributes and used to inform investment decisions. This approach enables targeting of underrepresented cohorts of interest for which limited data are available.

6 Limitations and future research

Several limitations should be acknowledged when interpreting the findings of this study. The analysis focuses on revealed route choices within a single regional city with relatively low cycling levels and a specific land use and network structure. Although we found behavioural patterns similar to those reported in other cities, the results of the route choice models are based on current bike riders in CoGB and on the existing street network. Care is therefore needed when applying these findings to contexts with different urban forms, traffic conditions, or cycling cultures. Given the relative ease of using PPGIS for data collection, future research could apply the same approach in other settings to examine route choice under varied conditions.

Although the set of route-level attributes was relatively rich, not all relevant influences on route choice were included. For example, we did not collect information on the time of day, so factors such as lighting or shade could not be accounted for. Similarly, we did not capture variation in weather conditions, which may shape riders' responses to slope, traffic stress, or canopy. Infrastructure variables reflected the presence of facilities rather than their quality or maintenance, even though surface condition, continuity, and upkeep are likely to affect usability. Including these dimensions and exploring within day and seasonal differences in route choice would be a valuable direction for future work.

The sample is another key limitation. We recruited participants primarily through social media and bicycle user group mailing lists, which are likely to overrepresent riders already engaged with cycling communities or local councils' communication channels. Completing an online, map-based survey is time-consuming, even with a small monetary incentive, so people with limited time or lower interest in the topic may be underrepresented. Participation was voluntary and online, introducing self-selection and potentially excluding people without reliable internet access or confidence in using digital maps. Routes were self-reported rather than tracked and may be affected by recall error, especially for infrequent or very short trips. Respondents may have prioritised their most salient or problematic routes rather than their complete set of journeys. These issues do not invalidate the results, but they indicate that PPGIS data should be interpreted as a critical window onto cycling behaviour rather than a complete record of all trips.

7 Conclusion

This study examined cyclists' route choice behaviour in the CoGB using map-based PPGIS data and a suite of PSL, MPSTL, and group-specific models. The results show that cyclists consistently avoid stressful traffic environments, steep slopes, and indirect routes, and prefer distinctive routes that differ meaningfully from available alternatives. The analysis highlights substantial heterogeneity in these preferences across gender, age, and trip purpose, and reveals context-specific roles for canopy cover and intersection density within a regional street network. The findings demonstrate that PPGIS can provide a practical and informative basis for route-level modelling in resource-constrained set-

tings and offer clear guidance for the design of continuous, direct, and low-stress cycling corridors that better serve diverse rider groups in regional cities.

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Appendix A: Survey interface and route-mapping

Figure 4 shows the map-based public participatory survey interface used to collect cycling route data. Respondents were presented with an interactive map and asked to select the trip purpose before marking their preferred cycling route for that destination. Clear instructions were provided to ensure consistency in route drawing, while allowing respondents flexibility to represent their preferred route.

Figure 5 illustrates an example of a route drawn by a participant for a specific trip purpose. The preferred cycling route is shown as a purple line overlaid on the map, created by clicking along the road network. These user-drawn routes formed the basis for subsequent network-matching and route-choice modelling.

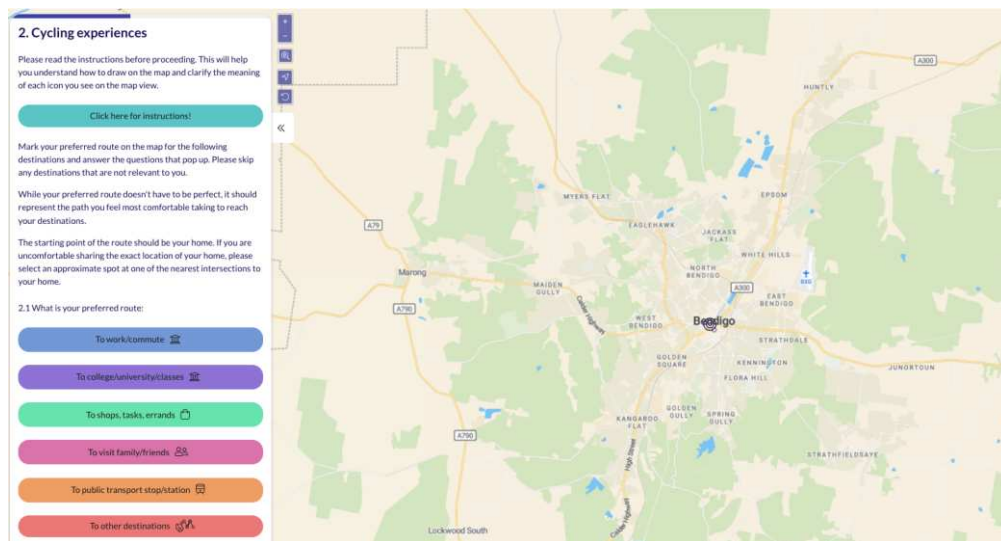


Figure 4: Map-based public participatory survey interface showing trip-purpose selection and instructions for drawing preferred cycling routes.

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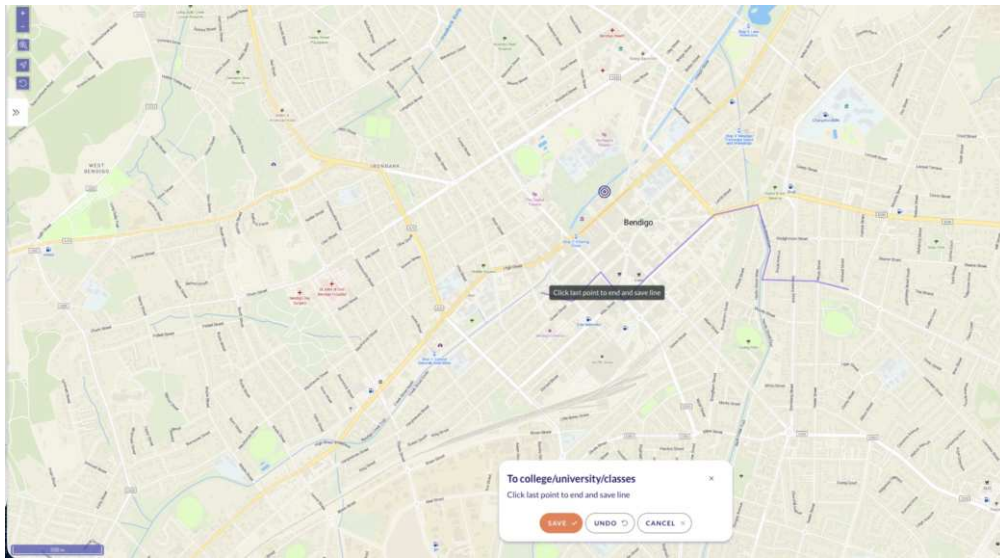


Figure 5: Example of a cycling route drawn by a survey participant, shown as a purple line on the map.

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